

Find the frequency counts for the following

```
for (i=1; i<n; i++);  
    printf("Hello");
```

→ 3 units

i=1; → 1

i=i+20; → 2

printf("%d", i); → 1

i=1; → 1

while (i<=n) → n+1

{

printf("%d", i); → n

i++; → n

}

→ 3n+2

For all the shell program done in os lab find out time complexity using frequency count

Addition of 2 numbers

echo -n "Enter 1st number" → 1

read n1 → 1

echo -n "Enter 2nd number" → 1

read n2 → 1

sum = \$((\$n1 + \$n2)) → 2

echo "sum is \$sum" → 1
7

Asymptotic Notation:

→ It is a general form of notation to calculate complexity

→ It is this general form (Notation) produces basic nature of the algorithm

Asymptotic notation of an algorithm is a mathematical representation of its complexity

NOTE: In asymptotic notation, when we want to represent the complexity of an algorithm, we use the most significant terms in the algorithm